FAST-NUECS

Spring 2026

Subject: ML for Robotics (CS-4090)

Instructor: Dr. Maryam Iqbal **Marks** (10)

Assignment# 04 (CLO-4) **Class:** BCS 8 A

Date: 4th May 2026 **Due Date:** 8th May 2026

Object Classification Task

Q1. Design and implement a machine learning solution for a real-world robotics problem, including data preprocessing, model selection, training, and validation.[CLO 4]

Problem Statement

In robotic systems, perception is critical for decision-making. Consider a mobile robot or robotic manipulator that must identify objects based on sensor data.

Your task is to design a classification-based machine learning system that enables a robot to classify objects into predefined categories.

Dataset

You may use:

- A **public dataset** (e.g., UCI ML Repository, Kaggle), OR
- A **synthetic dataset** simulating robotic sensor readings, OR
- A **self-collected dataset** (optional bonus)

Example features:

- Distance readings
- Object size
- Color intensity
- Weight
- Sensor voltage readings

Algorithms to Implement

a. K-Nearest Neighbors (KNN)
b. Naïve Bayes

Tasks

1. **Problem Formulation**

   Define the robotics problem, identify input features and output classes, and justify classification approach.

2. **Data Preprocessing**

   Handle missing values, normalize features, encode categorical data, and split dataset.

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3. **Model Implementation**

   Implement KNN and Naïve Bayes using appropriate libraries.

4. **Training and Validation**

   Evaluate using accuracy, confusion matrix, precision, recall, and F1-score.

5. **Performance Comparison**

   Compare both models and justify best choice.

6. **Robotics Context Integration**

   Explain real-world robotic application and constraints.

7. **Report & Code Submission**

   Submit report and well-commented code.